

PROJECT REPORT

**WhatNext Vision Motors:
Shaping the Future of Mobility with Innovation and
Excellence**

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Salesforce CRM Project

1. Project Overview

WhatNext Vision Motors is a Salesforce-based automotive management solution developed to improve the handling of vehicles, dealers, customers, orders, test drives and service requests. The project brings these activities together in a single system and uses automation to reduce repetitive manual work.

The ordering process is designed to improve customer convenience by identifying a suitable dealer using customer location information. The system also checks vehicle stock before an order is accepted, helping prevent orders for unavailable vehicles.

The solution also supports automated order-status processing. Orders affected by unavailable stock can remain pending, while orders for available vehicles can be confirmed. Scheduled processing can later review pending orders when stock becomes available.

2. Requirements

2.1 Salesforce CRM Implementation

- Store and manage vehicle details, stock availability and dealer information.
- Maintain customer orders, test drives and service requests.
- Automate assignment of orders to a suitable dealer based on customer location.

2.2 Process Automation

- Prevent order placement when the selected vehicle is out of stock.
- Assign orders to the appropriate dealer using customer location information.
- Send automated email reminders for scheduled test drives.

2.3 Apex and Trigger

- Use Apex triggers to enforce stock validation and order-related business rules.
- Use a trigger-handler class to keep the implementation organized and maintainable.

2.4 Batch Jobs

- Periodically check vehicle stock levels and process eligible orders.
- Support scheduled processing and notifications related to stock and orders.

3. Learning Outcomes

- Data modelling
- Fields and relationships
- Lightning App Builder
- Record-triggered flows
- Apex and Apex triggers
- Batch Apex
- Scheduled Apex

4. Developer Account Creation

The project begins by creating a Salesforce Developer organization. The registration requires basic user and organization information, followed by email verification and password setup.

Steps

- Open the Salesforce Developer signup page.
- Enter first name, last name, email, role, company, country, postal code and username.
- Submit the registration form.
- Open the verification email and activate the account.
- Set a password and security question.
- Enter the Salesforce setup environment after activation.

5. Objects and Relationships

Custom objects are used to represent the main entities in the automotive business process. Each object stores information for a specific area and relationships connect the records where necessary.

Object	Purpose	Main relationship
Vehicle__c	Stores vehicle details and stock.	Dealer and Orders
Vehicle_Dealer__c	Stores authorized dealer information.	Orders
Vehicle_Customer__c	Stores customer details.	Orders and Test Drives
Vehicle_Order__c	Tracks vehicle purchases.	Customer and Vehicle
Vehicle_Test_Drive__c	Tracks test-drive bookings.	Customer and Vehicle
Vehicle_Service_Request__c	Tracks vehicle servicing requests.	Customer and Vehicle

5.1 Vehicle Object

The Vehicle custom object provides a structured record for vehicle information. It is created from Object Manager using a text-based Vehicle Name record field. Reporting and search can be enabled for easier management.

5.2 Dealer Object

The Vehicle Dealer object stores detailed dealer information. Dealer Name is used as the text-based record name, with reporting and search enabled.

5.3 Remaining Objects

The customer, order, test-drive and service-request objects are created in the same manner so that the complete business process can be represented in Salesforce.

6. Tabs and Lightning Application

6.1 Custom Tabs

Custom tabs provide navigation to the objects created for the project. A Vehicle tab can be created from Setup by opening Tabs and selecting New under Custom Object Tabs. The same process is followed for the remaining custom objects.

6.2 Lightning Application

A Lightning application named WhatNext Vision Motors brings the project components together in one user interface.

- Open App Manager from Setup and create a new Lightning App.
- Enter the application name and description.
- Keep the standard application options unless a change is required.
- Add Vehicle, Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard to navigation.
- Add the System Administrator profile and save the application.

7. Fields and Relationships

7.1 Vehicle Fields

- Vehicle Name — Text
- Vehicle Model — Picklist
- Stock Quantity — Number
- Price — Currency
- Dealer — Lookup relationship
- Status — Picklist

7.2 Dealer Fields

- Dealer Name — Text
- Dealer Location — Text
- Dealer Code — Auto Number
- Phone — Phone
- Email — Email

7.3 Order Fields

- Vehicle Customer — Lookup
- Vehicle — Lookup
- Order Date — Date
- Status — Picklist

7.4 Customer, Test Drive and Service Fields

- Customer records include name, email, phone, address and preferred vehicle type.
- Test-drive records include customer, vehicle, date and status.
- Service-request records include customer, vehicle, service date, issue description and status.

7.5 Creating a Lookup Relationship

A lookup relationship is created on the Vehicle object to connect a vehicle with a dealer. This allows dealer information to be referenced from vehicle records and used by later automation.

8. Flow Automation

8.1 Automatic Dealer Assignment

A record-triggered flow is created for Vehicle Order records. When a new order is created with a Pending status, the flow retrieves the customer information and uses the customer's address to identify the relevant dealer. The order is then updated with the dealer information.

- Create a new record-triggered flow.
- Select Vehicle Order as the object.
- Run the flow when a record is created.
- Set the entry condition to Status = Pending.
- Get the related customer information.
- Find the relevant dealer using the customer address.
- Update the order with the dealer record.
- Save and activate the flow as Auto Assign Dealer.

8.2 Test Drive Reminder Flow

A second record-triggered flow handles reminders for scheduled test drives. It uses the test-drive date as the time source and creates a scheduled path one day before the appointment.

- Select Vehicle Test Drive as the flow object.
- Run the flow when a record is created or updated.
- Set the condition to Status = Scheduled.
- Create a scheduled path one day before Test_Drive_Date__c.
- Retrieve the customer's email information.
- Use a Send Email action to deliver the reminder.
- Save and activate the flow as Test Drive Reminder.

9. Apex and Trigger Processing

9.1 Trigger Handler

The Apex implementation uses a handler class named VehicleOrderTriggerHandler. The handler separates before-processing and after-processing logic so that stock validation and stock updates can be managed independently.

- Before insert or update: validate vehicle stock.
- After insert or update: update stock for confirmed orders.
- The trigger calls the handler class instead of containing all business logic directly.

9.2 Stock Validation

The handler collects the vehicle IDs from incoming orders, retrieves the corresponding stock quantities and checks whether sufficient stock exists. If a vehicle has no stock, the order receives an error and cannot be placed.

```
if (vehicle.Stock_Quantity__c <= 0) {  
    order.addError(  
        'This vehicle is out of stock. Order cannot be placed.'  
    );  
}
```

9.3 Stock Update

When an order is confirmed, the corresponding vehicle stock quantity is reduced by one. The changed vehicle records are then saved.

```
if (vehicle.Stock_Quantity__c > 0) {  
    vehicle.Stock_Quantity__c -= 1;  
    vehiclesToUpdate.add(vehicle);  
}  
  
if (!vehiclesToUpdate.isEmpty()) {  
    update vehiclesToUpdate;  
}
```

10. Batch Processing

Batch Apex is used to process pending vehicle orders in manageable groups. If a customer places an order while the selected vehicle is out of stock, the order can remain pending. After additional stock is added, a later batch run can review the order and update it when the required conditions are satisfied.

10.1 Batch Scheduler

A schedulable Apex class is used to start the batch job automatically. The project uses a batch size of 50 records.

```
global class VehicleOrderBatchScheduler implements Schedulable {
    global void execute(SchedulableContext sc) {
        VehicleOrderBatch batchJob = new VehicleOrderBatch();
        Database.executeBatch(batchJob, 50);
    }
}
```

10.2 Scheduling the Batch

The batch job can be scheduled to run automatically using a cron expression. The project demonstrates a daily schedule for vehicle-order processing.

```
String cronExp = '0 0 12 * * ?';

System.schedule(
    'Daily Vehicle Order Processing',
    cronExp,
    new VehicleOrderBatchScheduler()
);
```

11. End-to-End Process

- Vehicle, dealer and customer information is stored in Salesforce.
- A customer order is created for a selected vehicle.
- Apex validates the available stock.
- The flow assigns the appropriate dealer using customer location information.
- Scheduled test-drive reminders are sent automatically.
- Confirmed orders reduce the vehicle stock quantity.
- Scheduled batch processing reviews pending orders after stock changes.

12. Benefits

- Improves consistency in vehicle order processing.
- Reduces manual work through automation.
- Provides organized data through custom objects and relationships.
- Separates trigger logic using an Apex handler.
- Supports scalable processing through Batch Apex.
- Provides a single Lightning application for project operations.

13. Conclusion

The WhatNext Vision Motors project demonstrates the use of Salesforce to manage an automotive business process from vehicle and dealer data through customer orders and follow-up activities. Custom objects and relationships provide the data structure, while Lightning configuration creates the application interface.

Record-triggered flows automate dealer assignment and customer reminders. Apex and triggers enforce stock-related rules, while Batch Apex and scheduled execution support the processing of pending orders.

Together, these components provide a structured and automated solution for vehicle management and customer ordering.

Thank You